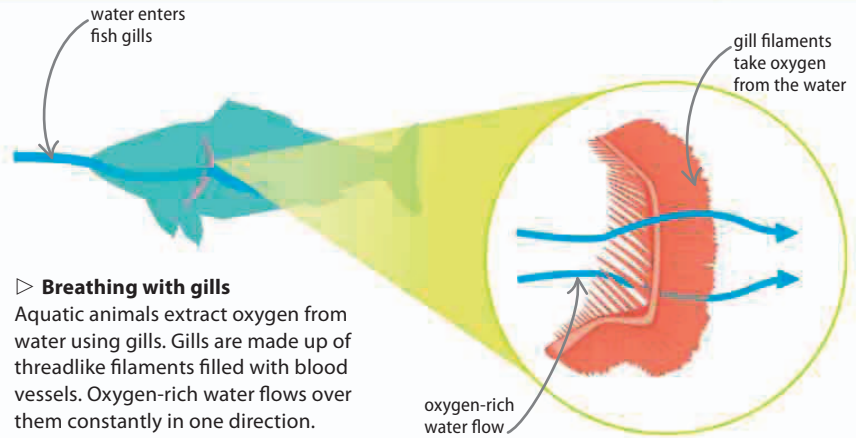


Gas exchange

Respiration requires a supply of oxygen, and the body also needs to remove the waste carbon dioxide it produces. The area through which these gases enter and leave the body is called the gas exchange surface. Lungs, gills, and the trachea tubes of insects are lined with these surfaces. A gas exchange surface is thin, moist, and well supplied with blood to take away the oxygen and deliver the waste carbon dioxide. The gases move in and out of the area by diffusion (see page 24).



▷ Breathing with gills

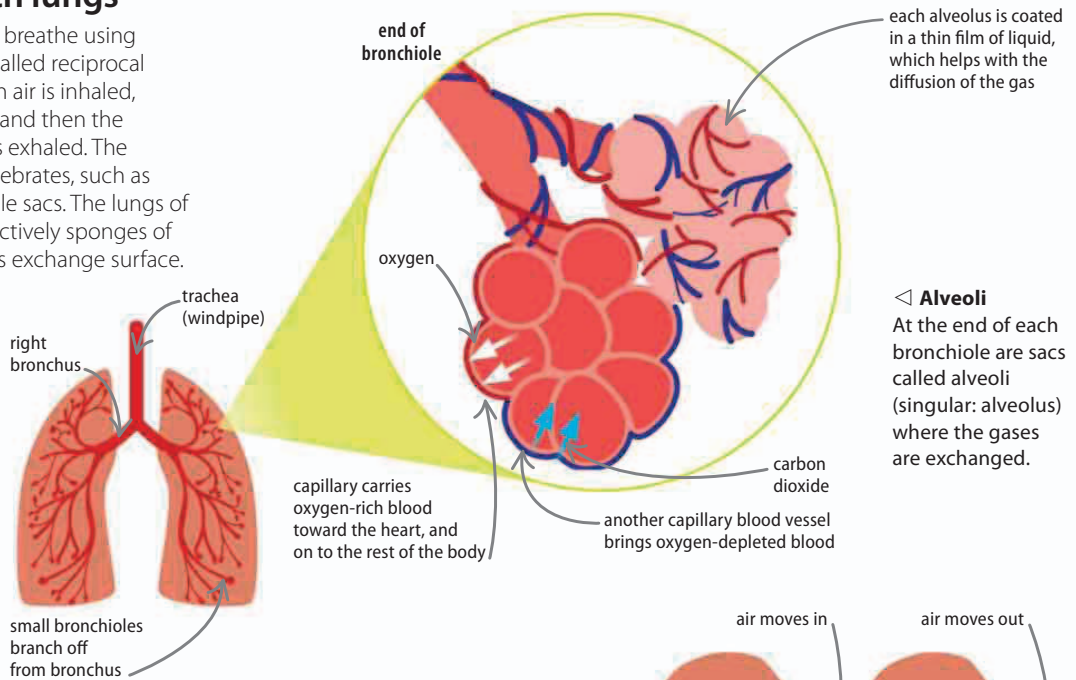
Aquatic animals extract oxygen from water using gills. Gills are made up of threadlike filaments filled with blood vessels. Oxygen-rich water flows over them constantly in one direction.

Breathing with lungs

Most land vertebrates breathe using lungs. The process is called reciprocal breathing: oxygen-rich air is inhaled, gases are exchanged, and then the oxygen-depleted air is exhaled. The lungs of primitive vertebrates, such as salamanders, are simple sacs. The lungs of larger animals are effectively sponges of tissue, with a huge gas exchange surface.

▷ Lungs

When you inhale, air is sucked into your lungs via your trachea, which branches into left and right bronchi, which in turn branch off into bronchioles.



◁ Alveoli

At the end of each bronchiole are sacs called alveoli (singular: alveolus) where the gases are exchanged.

▽ Gas mixture

The air we breathe is a mixture of gases. Only about a fifth of it is oxygen, which diffuses into the blood. There is about 100 times more carbon dioxide in exhaled air than in inhaled air.

Gas	Inhaled air %	Exhaled air %
nitrogen	78	78
oxygen	21	17
inert gas	1	1
carbon dioxide	0.04	4
water vapor	little	saturated

▷ Reciprocal breathing

To breathe in, the diaphragm moves down, enlarging the space in the chest. This lowers the pressure in the lungs, forcing in air from outside. To breathe out, the diaphragm goes up, reducing the space in the chest and pushing out the oxygen-depleted air.

